Our customers include service providers, such as:

A1 Bulgaria
A1 Telekom Austria Group
ABN Amro Bank NV
Airtel
Altice France
Altice USA
América Móvil
AT&T
AT&T Mexico
Azercell
Bank Hapoalim
Beeline
Bell Canada
Bharat Sanchar Nigam Limited (BSNL)
Botswana Telecommunications Corporation
BT
Cable & Wireless
Capita
Cellcom
Charter Communications
Claro Argentina
Claro Brasil
Claro Chile
Claro Dominican Republic
Claro Puerto Rico
Colt Technology Services
Comcast
Consolidated Communications
DELTA Fiber
Deutsche Telekom
Dish
DTV
EE
Elisa
etisalat by&e
Far EasTone
Fastweb
Flow
Foxtel
Frontier Communications
Globe Telecom
J:COM
KT
Kyivstar
LG Uplus
Liberty Global
Lumen
M1
Magyar Telekom
Maxis
Melon Digital
Merto Fibernet
MGM+
MTS
NTT Infrastructure Network Corporation
Optus
Orange Belgium
Orange Liberia
Orange Spain
Paramount
Partner
PLDT
PPF Telecom Group
Proximus
Rogers
Safaricom
SES
Singtel
Sky Italia
Sky UK
StarHub
Sunrise
Telefónica Argentina (Movistar)
Telefónica Brasil (Vivo)
Telefónica Chile (Movistar)
Telefónica Peru (Movistar)
Telenet
Telia Norway
Telia Sweden
Telkom SA
Telkomsel
Telstra
TELUS
Three Ireland
Three UK
Thryv
TIM Brazil
TPG Telecom
True
Turner
T-Mobile
UPC Broadband
US Cellular
VEON
Verizon
Virgin Media
Vodacom
Vodafone Albania
Vodafone Czech Republic
Vodafone Germany
Vodafone Hungary
Vodafone Idea
Vodafone Ireland
Vodafone Italy
Vodafone Romania
Vodafone Spain
Vodafone Turkey
Vodafone UK
VodafoneZiggo
Warner Bros
Wind Tre
Winity Telecom
XL Axiata

The following is a summary of revenue by geographic area. Revenue is attributed to geographic region based on the location of the customer:

	Year Ended September 30,		
	2024	2023	2022
North America	66.5%	67.7%	67.8%
Europe	14.5%	14.4%	12.7%
Rest of the World	19.0%	17.9%	19.5%

Competition

The market for customer experience solutions in the communications and media industry continues to be highly competitive. Amdocs' competitive landscape is comprised of internal IT departments of our customers, as well as independent competitors or new entrants that may compete broadly with us or in limited segments of our market, and can be generally categorized as follows (competitors in each category referenced below in alphabetical order):

- providers of BSS/OSS and customer relationship management (CRM)/digital systems, including CSG International, Netcracker (a NEC subsidiary), Optiva, Oracle, Pegasystems, Salesforce, SAP and ServiceNow;
- system integrators and providers of IT services, such as Accenture, Cap Gemini, Cognizant, DXC Technology, Infosys, Kyndryl, Tata Consultancy Services, Tech Mahindra and Wipro (some of whom we also cooperate with in certain opportunities and projects);
- network equipment providers such as Ciena, Ericsson, Huawei, and Nokia, (some of whom we also cooperate with in certain opportunities and projects and some of whom also have BSS/OSS offerings); and
- niche domain players, often start-up companies, which compete against particular parts of our portfolio, such as Matrixx in charging; Hansen in catalog; Aria Systems, Stripe, Zuora in subscription billing; Deluxe Media and iNDEMAND in media; Slalom in cloud consulting; Material+ and Work & Co in experience design.

We expect the competition in our industry to increase from many of such companies.

We believe that we are able to differentiate ourselves from these competitors by, among other things:

- applying our 40-year heritage to the development and delivery of products and professional services that enable our customers to overcome their challenges and achieve service differentiation by migrating to the cloud, providing a personalized and intelligent customer experience, shaping the quality of network experience and simplifying the complexity of the operating environment;
- continuing to design and develop solutions targeted specifically to the communications and media industry;
- innovating and enabling our customers to quickly adopt new business models that will improve their ability to drive new revenues, and compete and win in a changing market;
- providing high-availability, high-quality, reliable, scalable, integrated and modular applications, leveraging cloud technology, AI, GenAI and other new software development and deployment options;
- providing flexible and tailored IT business process outsourcing solutions and delivery models; and
- offering customers end-to-end accountability from a single vendor.

Employees

We invest significant resources in the training, retention and motivation of high-quality personnel. Training programs cover areas such as technology, applications, development methodology, project methodology, programming standards, industry background, business, management development and leadership. Our management development efforts are reinforced by an organizational structure that provides opportunities for talented managers to gain experience in general management roles. We also invest considerable resources in personnel motivation, including providing various incentive plans for sales staff and high-quality employees. Our future success depends in large part upon our continuing ability to attract and retain highly qualified managerial, technical, sales and marketing personnel and outstanding leaders. Moreover, we are committed to diversity and inclusion, believing a gender-diverse, multi-cultural workforce spread across the globe provides strength and a competitive business advantage.

See “Directors, Senior Management and Employees — Workforce Personnel” for further details regarding our employees and our relationships with them.

Property, Plants and Equipment

Our principal capital expenditures for fiscal years 2024, 2023 and 2022 have been for computer equipment and software in our operating facilities and development centers, for which we spent approximately $72 million, $84 million and $92 million, respectively, and for the development of our new campus in Israel, for which we spent approximately $116 million in fiscal year 2022, (the amounts for fiscal years 2023 and 2024 were immaterial as we completed the move into our new campus during fiscal year 2023).

Facilities

Our properties consist of leased and owned (only in Ra’anana, Israel) facilities an aggregate of approximately 2.9 million square feet worldwide, including significant leases in the Canada, Cyprus, India, Israel, the Philippines, Mexico, the United Kingdom and the United States. The following table summarizes information with respect to the principal facilities leased and owned by us and our subsidiaries as of September 30, 2024:

Location	Area (Sq. Feet)
Americas	543,116
EMEA	1,076,941
APAC	1,311,920
Total	2,931,977

Our leases expire on various dates through 2033. In fiscal year 2023, the Company has started to use its campus in Ra’anana, Israel on land acquired by a legal entity owned equally by the Company and Union Investments and Development Limited (“Union”) pursuant to agreements entered into by the Company and Union in December 2017. The campus provides an advanced working environment that meets the needs of Amdocs Israel and its employees, and supports the Company’s future growth. The design for the campus is in accordance with LEED Gold standards and includes advanced energy and water saving systems.

Equipment

We develop our solutions over a system of Linux and Windows servers owned or leased by us, as well as over cloud providers. We use a variety of software products in our development centers, including products by Microsoft, CouchBase, Syncsort, Red Hat, Oracle, CA, IBM, Hewlett-Packard or others. Our data storage is based mainly on equipment from EMC, InfiniDat, IBM and Hewlett-Packard.

ITEM 4A. UNRESOLVED STAFF COMMENTS

Not applicable.

ITEM 5. OPERATING AND FINANCIAL REVIEW AND PROSPECTS

Overview of Business and Trend Information

Amdocs is a leading provider of software and services for communications and media industry service providers in both developed countries and emerging markets. We believe the demand for our solutions is driven by our customers’ continued migration to the cloud, deployment of 5G standalone, fixed wireless access (FWA), and fiber networks and their transformation into digital service providers to provide connectivity services, content and applications (apps) on any device through digital and non-digital channels. Regardless of whether service providers are bringing their first offerings to market, scaling for growth, consolidating systems or transforming the way they do business, we believe that they seek to differentiate their offerings by delivering a customer experience that is simple, personal, contextual and valuable at every point of engagement and across all channels.

We believe service providers will maintain a strong focus on growing new revenue streams, cost reduction and driving more efficient operations, and that the trends of ongoing digital transformation with a focus on customer experience, migration to the cloud, next-generation networks, and consolidation within the industry will continue. Service providers are increasingly focusing on their core capabilities and investing in 5G and fiber rollouts to meet the demand for increased bandwidth, faster pace of innovation for new digital services and introducing GenAI, as well as to improve their business and operational agility and optimize and monetize their investments in such services. At the same time, many service providers are partnering with leading suppliers to offer their customers a rich portfolio of offerings including media; entertainment; enterprise enablement; Internet of Things (IoT); and digital lifestyle services, all of which is driving growth in the demand for multi-modal customer engagement capabilities, data, and GenAI.

We develop, implement and manage software and services designed to meet our customers' business needs and empower them to transform their boldest ideas into reality. Our technology, design-led thinking approach and expertise help service providers to migrate to the cloud, manage and monetize their next-generation networks, further transform into digital service providers, accelerate their GenAI journeys, enhance their entertainment offerings, and serve their customers across all channels. With our portfolio's open and modular structure, organized by capability and matched to industry standards, our customers have the flexibility to choose business offerings that address their specific needs and improve their time to market and value.

In part, we have sought, through acquisitions, to expand our products and services offerings and customer base and to enhance our ability to provide managed services to our customers. In recent years, we have completed numerous acquisitions (including our fiscal year 2022 acquisitions of Roam Digital and DevOpsGroup, our fiscal year 2023 acquisitions of the service assurance business of TEOCO, and of ProCom Consulting and our fiscal year 2024 acquisitions of Astadia, and the network engineering businesses of Pramira), which, among other things, we believe will enable us to expand our 5G, fiber, digital and cloud-native capabilities, service assurance and cloud migration expertise. As part of our strategy, we may continue to pursue acquisitions and other initiatives in order to offer new products or services, enter into new vertical markets or otherwise enhance our market position or strategic strengths.

In evaluating Amdocs' portfolio of products, services and business activities in relation to our strategic investment priorities for fiscal year 2025, we decided to phase out several low-margin, non-core business activities that we believe are becoming commoditized and hold little potential for long-term value addition or profitability enhancement in the future. These activities, which generated revenue of approximately $600 million in fiscal year 2024, included, among others, certain low-margin software and hardware partner activities, Vubiquity's transactional video on demand business and other non-core subscription services. These activities were substantially already ceased in the first quarter of fiscal 2025.

The Amdocs Offerings

Our portfolio consists of software and services that address service providers' business and operational needs. Our offerings, grouped by technology capabilities such as monetization, commerce and care, service and network automation, are designed to meet the challenges facing our customers as they roll out 5G networks, migrate to the cloud, introduce GenAI solutions, and transform into digital service providers within the framework of a hybrid IT environment, which requires them to rapidly introduce new cloud-native applications while still operating legacy systems. Our software is designed to enable modular expansion as a service provider evolves, and its microservices-based architecture enables the rapid deployment of complex applications as suites of independently deployable services that can be frequently upgraded via DevSecOps.

Our comprehensive line of services is designed to address every stage of a service provider's lifecycle. They include consulting, experience design, data, cloud, network services, to delivery, quality engineering (testing), operations, systems integration, and content services. Our managed services provide multi-year, flexible and tailored support, managing IT business processes and applications services. They include application development, modernization and maintenance, IT and infrastructure services, testing and professional services that are designed to assist customers in the selection, implementation, operation, management and maintenance of their IT systems.

We believe that our business model of developing mission-critical software, deploying it at our customers and then operating it and supporting it on an ongoing basis, provides Amdocs with a high level of recurring revenue. This, together with our scalable global resource allocation model and our continuous operational excellence, automation and efficiency improvements, enables us to deliver consistent operating margin performance over time.

We conduct our business globally, and as a result we are subject to the effects of global economic conditions and, in particular, market conditions in the communications and media industry. In fiscal year 2024, customers in North America accounted for 66.5% of our revenue, while customers in Europe and the rest of the world accounted for 14.5% and 19.0%, respectively. We maintain development facilities in Brazil, Canada, Cyprus, India, Ireland, Israel, Mexico, the Philippines, the United Kingdom and the United States. Historically, AT&T has been our largest customer, accounting for 24.5% and 23.8% of our revenue in fiscal years 2024 and 2023, respectively. In fiscal years 2024 and 2023, our next largest customer, T-Mobile, accounted for 22.6% and 23.1% of our revenue, respectively. Aggregate revenue derived from the multiple business arrangements we have with our ten largest customers accounted for approximately 70% of our revenue in fiscal years 2024 and 2023. We believe that demand for our solutions is primarily driven by the following key factors:

- Transformation within the communications and media industry, including:
  - continued transformation of service providers to digital service providers;
  - service provider migration to the cloud;
  - increasing use of communications and content services;

    - widespread access to content, information and applications;
    - expansion into new lines of business;
    - consolidation among service providers in established markets, often including companies with multinational operations;
    - increased competition, including from non-traditional players;
    - continued bundling and blending of communications and entertainment; and
    - continued commoditization and pricing pressure.
- Technology advances, such as:
    - wide-scale foundational technology changes, including the leveraging of open-source, cloud-enabled and cloud-native operating infrastructure, microservices-based architecture, API-based ecosystems, and aggressive digital modernization transformations;
    - evolving service provider business models and opportunities like OTT partnerships, content development and offerings, enterprise and small or medium-sized business modernization, and innovative consumer bundling solutions;
    - network evolution in order to support growing technology needs associated with Internet of Things (IoT), autonomous vehicles and augmented and virtual reality;
    - new communications technologies such as 6G wireless technology, 5G wireless technology, fixed wireless access, eSIM, Wi-Fi 6, and Narrowband IoT (NB-IOT),
    - adoption of AI, including GenAI, ML, robotic process automation (RPA), and natural language processing (NLP) edge computing, network and service automation, and blockchain, and;
    - additional evolution of technology in select domains, including edge computing, quantum computing, serverless compute and blockchain among others.
- Customer focus, such as:
    - the need for service providers to personalize the customer's experience and provide contextual and personalized engagements at all points in their omni-channel customer journey;
    - increasing customer expectations for new, innovative services and applications that are personally relevant and that can be accessed anytime, anywhere and from any device;
    - the ever-increasing expectations for service and support, including omni-monetization and proactive multi-modal customer care and commerce; and
    - continuous proliferation of on-demand experiences, including low-latency, high quality of service connectivity and seamless digital interactions.
- The need for operational efficiency, including:
    - the shift from in-house management to vendor solutions;
    - business needs of service providers to reduce costs and lower total cost of ownership of software systems while retaining high-value customers in a highly competitive environment;
    - automating, introducing GenAI, and integrating business processes that span service providers' business systems and network solutions;
    - leveraging GenAI to optimize employee processes, simplify development of systems, and drive agility in the resolution of operational issues;
    - implementing and integrating new next-generation networks (and retiring legacy networks) to deploy new technologies; and
    - transforming fragmented legacy OSS to introduce new, orchestrated and automated services in a timely and cost-effective manner.

Revenue from managed services arrangements is a significant part of our business, generating substantial, long-term recurring revenue streams and cash flow. Revenue from managed services arrangements accounted for approximately $2.90 billion and $2.86 billion of revenue in fiscal years 2024 and 2023, respectively. In managed services contracts, revenue from the operation of a customer's system is recognized as services are performed based on time elapsed, output produced or volume of data processed. In the initial period of our managed services projects, we often invest in modernization and consolidation of the customer's systems and may see also additional modernization cycles in our more mature managed services engagements. Managed services engagements can be less profitable in their early stages; however, margins tend to improve over time, and this improvement is seen more rapidly in the initial period of an engagement, as we derive benefit from insertion of automation tools, AI, GenAI, operational efficiencies and from changes in the geographical mix of our resources.

Research and Development, Patents and Licenses

Our research and development activities involve the development of new software architecture, modules and product offerings in response to an identified market demand. We also expend additional amounts on applied research and software development activities to keep abreast of new technologies in the communications and media industry and to provide new and enhanced functionality to our existing product offerings. We leverage leading-edge development technologies and associated technologies, for example, GenAI, DevSecOps, CI/CD and Agile, to ensure we are able to develop and deliver our solutions efficiently and cost-effectively.

Substantially all of our research and development expenditures are directed at our solutions. In recent years, we have also invested our research and development efforts in network control, optimization and orchestration and network functions virtualization technologies; applications to enable service providers to deploy and monetize technologies such as fiber, LTE, 5G, small cells and Wi-Fi; big data analytics and intelligence capabilities leveraging AI, GenAI and NLP toward consumer and business satisfaction, marketing effectiveness and network operations and experience; increased focus for the business segment, digital, commerce and entertainment domains; platforms for processing, distributing and monetizing content globally and on foundational technologies including microservices and all major cloud providers. We believe that our research and development efforts are a key element of our strategy and are essential to our success. However, an increase or a decrease in our total revenue would not necessarily result in a proportional increase or decrease in the levels of our research and development expenditures, which could affect our operating margin.

Our products are largely comprised of software and systems that we have developed or acquired and that we regard as proprietary. In recent years, we have invested in adopting open source components in an effort to reduce total cost of ownership for our customers, but our software and software systems remain the results of long, robust and intensive development processes. Although our technology is not significantly dependent on patents or licenses from third parties, certain aspects of our products continue to make use of software components licensed from third parties. As a developer of complex software systems, third parties may claim that portions of our systems infringe their intellectual property rights. The ability to develop and use our software and software systems requires knowledge and professional experience that we believe would be very difficult for others to independently obtain. However, our competitors may independently develop technologies that are substantially equivalent or superior to ours. We have taken, and intend to continue to take, several measures to establish and protect our proprietary rights in our products and technologies from third-party infringement. We rely upon a combination of trademarks, patents, contractual rights, trade secret law, copyrights and non-disclosure agreements. We enter into non-disclosure and confidentiality agreements with our customers, employees and marketing representatives and with certain contractors with access to sensitive information; and we also limit customer access to the source code of our software and software systems.

Operating Results

The following table sets forth for the fiscal years ended September 30, 2024, 2023 and 2022, certain items in our consolidated statements of income reflected as a percentage of revenue (figures may not sum because of rounding):

	Year Ended September 30,		
	2024	2023	2022
Revenue	100%	100%	100%
Operating expenses:			
Cost of revenue	64.9	64.7	64.6
Research and development	7.2	7.7	7.8
Selling, general and administrative	11.4	11.7	11.5
Amortization of purchased intangible assets and other	1.2	1.2	1.6
Restructuring charges	2.6	1.5	—
	87.4	86.6	85.5
Operating income	12.6	13.4	14.5
Interest and other expense, net	(0.7)	(0.4)	(0.6)
Gain from sale of a business	—	—	0.2
Income before income taxes	11.8	13.0	14.2
Income taxes	1.9	1.9	2.2
Net income	9.9%	11.1%	12.0%
Net income attributable to noncontrolling interests	0.06	0.05	—
Net income attributable to Amdocs Limited	9.9%	11.1%	12.0%

Fiscal Years Ended September 30, 2024 and 2023

The following is a tabular presentation of our results of operations for the fiscal year ended September 30, 2024, compared to the fiscal year ended September 30, 2023. Following the table is a discussion and analysis of our business and results of operations for these fiscal years.

	Year Ended September 30,		Increase (Decrease)	
	2024	2023	Amount	%
	(In thousands)			
Revenue(1)	$ 5,004,989	$ 4,887,550	$ 117,439	2.4%
Operating expenses:				
Cost of revenue	3,249,598	3,159,941	89,657	2.8
Research and development	360,798	374,855	(14,057)	(3.7)
Selling, general and administrative	572,845	570,707	2,138	0.4
Amortization of purchased intangible assets and other	62,052	57,156	4,896	8.6
Restructuring Charges	131,088	70,901	60,187	84.9
	4,376,381	4,233,560	142,821	3.4
Operating income	628,608	653,990	(25,382)	(3.9)
Interest and other expense, net	(37,537)	(17,629)	(19,908)	112.9
Income before income taxes	591,071	636,361	(45,290)	(7.1)
Income taxes	94,750	93,399	1,351	1.4
Net income	$ 496,321	$ 542,962	$ (46,641)	(8.6)%
Net income attributable to noncontrolling interests	3,124	2,253	871	38.7
Net income attributable to Amdocs Limited	$ 493,197	$ 540,709	$ (47,512)	(8.8)%

(1) Geographic Information:

	Year Ended September 30,		Increase (Decrease)	
	2024	2023	Amount	%
	(In thousands)			
North America (mainly United States)	$ 3,325,967	$ 3,306,988	$ 18,979	0.6%
Europe	726,226	703,141	23,085	3.3
Rest of the world	952,796	877,421	75,375	8.6
Revenue	$ 5,004,989	$ 4,887,550	$ 117,439	2.4%

Revenue. Revenue increased by $117.4 million, or 2.4%, to $5,005.0 million in fiscal year 2024, from $4,887.6 million in fiscal year 2023. The increase in revenue for fiscal year ended September 30, 2024, was attributable to increased activity across all regions. Revenue for fiscal year 2024 increased by 2.7% compared to fiscal year 2023, excluding approximately 0.3% negative foreign currency fluctuations impact. Revenue from our two largest customers increased in aggregate by 3.0 % in fiscal year 2024 compared to fiscal year 2023, while revenue from all other customers also increased in aggregate by 2.5%, excluding approximately 0.6% negative foreign currency fluctuations impact.

In fiscal year 2024, revenue from customers in North America, Europe and the rest of the world accounted for 66.5%, 14.5% and 19.0%, respectively, of total revenue, compared to 67.7%, 14.4% and 17.9%, respectively, in fiscal year 2023.

Revenue from customers in North America increased in absolute amount in fiscal year 2024, primarily attributable to higher revenue from key customers in North America. Revenue from customers in North America increased in fiscal year 2024, while total revenue increased at a higher rate, primarily due to slower pipeline to sales conversion, which resulted in a decrease of revenue from customers in North America as a percentage of total revenue.

Revenue from customers in Europe as a percentage of total revenue remains stable. Revenue from customers in Europe increased in absolute amount in fiscal year 2024, as a result of an increase in managed services arrangements, as we expand our presence in this region.

Revenue from customers in rest of the world increased significantly in fiscal year 2024, as project and managed services activities continued to ramp up with various customers, primarily in the Asia-Pacific region.

Cost of Revenue. Cost of revenue consists primarily of costs associated with providing services to customers, including compensation expense and costs of third-party products and services, as well as fee and royalty payments to software suppliers. Cost of revenue increased by $89.7 million, or 2.8%, to $3,249.6 million in fiscal year 2024, from $3,159.9 million in fiscal year 2023. The cost of revenue as a percentage of total revenue slightly increased, from 64.7% in fiscal year 2023 to 64.9% in fiscal year 2024. The slight increase in cost of revenue as a percentage of revenue, was commensurate with revenue growth. Our cost of revenue was positively impacted by our continued focus on operational efficiency improvements and by foreign exchange fluctuations.

Research and Development. Research and development expense is primarily comprised of compensation expense. Research and development expense decreased by $14.1 million, or 3.7%, to $360.8 million in fiscal year 2024, from $374.9 million in fiscal year 2023, and decreased as a percentage of total revenue from 7.7% in fiscal year 2023, to 7.2% in fiscal year 2024. The decrease is attributable to accelerated research and development investments in prior periods. We continue to invest in our cloud offerings, 5G and network related innovation, AI and GenAI capabilities and further developing our digital offerings. Our research and development efforts are a key element of our strategy and are essential to our success, and we intend to maintain our commitment to research and development. However, increase or decrease in our revenue would not necessarily result in a proportional increase or decrease in the levels of our research and development expenditures, which could affect our operating margin. Please see “Research and Development, Patents and Licenses.”

Selling, General and Administrative. Selling, general and administrative expense, which is primarily comprised of compensation expense, slightly increased by $2.1 million, or 0.4%, to $572.8 million in fiscal year 2024, from $570.7 million in fiscal year 2023. Selling, general and administrative expense decreased as a percentage of total revenue from 11.7% in fiscal year 2023, to 11.4% in fiscal year 2024. This decrease was primarily attributable to costs reduction attributable to a focus on operational excellence through disciplined resource management and automation to drive efficiency improvements, which were partially offset by an increase in the account receivable allowances. Selling, general and administrative expense may fluctuate from time to time, depending upon such factors as changes in our workforce and sales efforts and the results of any operational efficiency programs that we may undertake.

Amortization of Purchased Intangible Assets and Other. Amortization of purchased intangible assets and other increased by $4.9 million, or 8.6%, to $62.1 million in fiscal year 2024, from $57.2 million in fiscal year 2023. The increase in amortization of purchased intangible assets and other was primarily attributable to an increase in amortization of intangible assets due to recent completed acquisitions, partially offset by a completion of amortization of previously purchased intangible assets and a decrease in acquisition-related costs.

Restructuring Charges. Restructuring charges in fiscal year 2024 were $131.1 million, increased by $60.2, from $70.9 in fiscal year 2023. The increase was due to actions taken under a new restructuring plan in fiscal year 2024, mainly comprised of employee's severance expense and benefits arrangements. The Company expects to execute the remainder of the 2024 restructuring plan over the next several quarters. For the restructuring charges for the comparable period and for more details, please see Note 10 to our consolidated financial statements.

Operating Income. Operating income decreased by $25.4 million, or 3.9%, to $628.6 million in fiscal year 2024, from $654.0 million in fiscal year 2023. Operating income decreased as a percentage of total revenue, from 13.4% in fiscal year 2023 to 12.6% in fiscal year 2024. The decrease in operating income was attributable primarily to an increase in restructuring charges recorded in fiscal year 2024 compared to fiscal year 2023, as discussed above. Excluding the restructuring charges of $131.1 and $70.9 million recorded in fiscal years 2024 and 2023, respectively, our operating income as a percentage of revenue would have increased by 60 basis points, primarily as a result of continued focus on operational excellence through disciplined resource management, automation, and tools leveraging AI as well as GenAI, to drive additional cost savings and efficiency improvements. Our operating income was positively affected by foreign exchange impacts.

Interest and Other Expense, Net. Interest and other expense, net, changed from a net expense of $17.6 million in fiscal year 2023 to a net expense of $37.5 million in fiscal year 2024. The change was primarily attributable to increase in interest expenses net of interest income, as a result of lower level of cash balances, changes in minority equity investments measured at fair value, and to a lesser extent, an increase in adverse foreign exchange fluctuation charges.

Income Taxes. Income taxes for fiscal year 2024 were $94.8 million on pre-tax income of $591.1 million, resulting in an effective tax rate of 16.0% in fiscal year 2024, compared to 14.7% in fiscal year 2023. Our effective tax rate may fluctuate between periods as a result of discrete items that may affect a particular period, please see Note 11 to our consolidated financial statements.

Net income attributable to Amdocs Limited. Net income decreased by $47.5 million, or 8.8%, to $493.2 million in fiscal year 2024, from $540.7 million in fiscal year 2023. The decrease in net income is primarily attributable to a decrease in operating income, primarily as a result of an increase in restructuring charges and an increase in interest and other expense, net.

Diluted Earnings Per Share. Diluted earnings per share decreased by $0.24, or 5.3%, to $4.25 in fiscal year 2024, from $4.49 in fiscal year 2023. The decrease in diluted earnings per share was attributable to a decrease in net income, partially offset by a decrease in the diluted weighted average number of shares outstanding, which resulted from share repurchases. The restructuring charges of $131.1 million and $70.9 million in fiscal years 2024 and 2023, decreased the diluted earnings per share by $0.98 and $0.46, respectively. Please see also Note 21 to our consolidated financial statements.

Results of Operations for Fiscal year 2023 Compared to Fiscal year 2022

For a comparative discussion and analysis related to the results of operations and changes in financial condition for fiscal year 2023 compared to 2022, refer to Item 5. "Operating and Financial Review and Prospects" in our Annual Report on Form 20-F for the fiscal year ended September 30, 2023, filed on December 13, 2023, with the SEC and available at www.sec.gov.

Liquidity and Capital Resources

Cash, Cash Equivalents and Short-Term Interest-Bearing Investments. Cash, cash equivalents and short-term interest-bearing investments, totaled $514.3 million as of September 30, 2024, compared to $742.5 million as of September 30, 2023. The decrease was mainly attributable to $563.1 million used to repurchase our ordinary shares, $212.0 million of cash dividend payments, $105.5 million for capital expenditures, net, $86.8 million of payments for business acquisitions, partially offset by $724.4 million in positive cash flow from operations, reflecting healthy cash collections and $26.9 million of proceeds from stock option exercises. Net cash provided by operating activities amounted to $724.4 million and $822.6 million in fiscal years 2024 and 2023, respectively.

Our free cash flow for fiscal year 2024 was $618.9 million and is calculated as net cash provided by operating activities of $724.4 million for the period less $105.5 million for capital expenditures, net, and after restructuring payments of $75 million.

Free cash flow is a non-GAAP financial measure and is not prepared in accordance with, and is not an alternative for, generally accepted accounting principles and may be different from non-GAAP financial measures with similar names used by other companies. Non-GAAP measures such as free cash flow should only be reviewed in conjunction with the corresponding GAAP measures. We believe that free cash flow, when used in conjunction with the corresponding GAAP measure, provides useful information to investors and management relating to the amount of cash generated by the Company's business operations.

We believe that our current cash balances, cash generated from operations, our current lines of credit, loans, Senior Notes and our ability to access capital markets will provide sufficient resources to meet our operational needs, loan and debt repayment needs, fund share repurchases and the payment of cash dividends for at least the next fiscal year.

We have short-term interest-bearing investments comprised of marketable securities and bank deposits. We classify all of our marketable securities as available-for-sale securities. Such marketable securities consist primarily of money market funds, corporate bonds, U.S. government treasuries and supranational and sovereign debt, which are stated at market value. We believe we have